3. Requirements Gathering

TawzeefAI – HR & Recruitment Screening Assistant

TawzeefAI is an AI-powered assistant that helps an HR team in an Egyptian company screen résumés against real job requirements, match candidates to open roles, answer employee questions about HR policy, and schedule interviews. It serves three groups of users (candidates, HR staff, and current employees) and keeps each group's data and permissions strictly separated. This section defines who the stakeholders are, what they need, and what the system must do and how well it must do it.

3.1 Stakeholder Analysis

3.1.1 Stakeholders and Their Needs

Stakeholder	Description	Needs
Candidate (external, primary)	A job applicant outside the company who interacts with TawzeefAI through the chat interface. Has access only to public job information and to his or her own application.	• Learn about open roles: responsibilities, requirements, and the hiring process. • Submit a résumé and apply for a role. • Check the status of his or her own application at any time. • Confirm or reschedule an interview slot. • Be confident that personal data is private and not visible to other candidates. • Receive clear, polite answers, including a clear message when the assistant cannot help.
HR Recruiter (internal, primary)	HR staff responsible for running the recruitment process and using TawzeefAI to reduce manual screening work.	• Define and maintain job roles and their requirements. • Screen résumés and match candidates to roles, with ranked results. • Understand why a candidate received a score (explainable scoring). • Track each candidate's status across the whole process. • Schedule interviews with minimal back-and-forth. • See recruitment metrics and the candidate pipeline on a dashboard. • Keep the final hiring decision in human hands.
Current Employee (internal, primary)	An existing employee who uses TawzeefAI to ask questions about company HR policies and procedures.	• Accurate answers to HR policy questions. • To see which official document the answer came from. • Quick answers without contacting HR for every question. • To be told honestly when no answer is available, instead of receiving a wrong one.
HR Manager (internal, secondary)	Senior HR staff who oversee the process, maintain the knowledge base, and are accountable for screening fairness.	• Review the audit log of screening decisions. • Monitor metrics such as time-to-hire and screening time saved. • Upload and update HR policy documents and job descriptions so answers stay current.
System Administrator / Project Team (technical, secondary)	The people who deploy and maintain TawzeefAI.	• Secure management of secrets (API keys, credentials). • Reliable deployment, clear error handling, and basic monitoring.

3.1.2 Role Access Overview

The table below defines what each user role may do. It is the basis for the Role-Based Access Control requirements (FR-02, NFR-01, NFR-02) and for the access-control tests.

Capability / Data	Candidate	Current Employee	HR Recruiter	HR Manager
View public job information and role requirements	Yes	Yes	Yes	Yes
Submit a résumé / apply for a role	Yes (own)	No	Yes (on behalf of candidates)	Yes
View application status	Own only	No	All candidates	All candidates
Confirm / reschedule interview slot	Own only	No	Manage all	Manage all
View other candidates' data	No	No	Yes	Yes
View match scores and explanations	No	No	Yes	Yes
Create / edit job roles	No	No	Yes	Yes
Ask HR policy questions (employee chat)	No	Yes	Yes	Yes
View recruitment dashboard	No	No	Yes	Yes
View screening audit log	No	No	No	Yes
Upload / update policy and job documents	No	No	No	Yes

Note: HR Recruiters and HR Managers are both "HR staff". Their views differ only by the extra Manager capabilities above.

3.2 User Stories & Use Cases

3.2.1 User Stories

Each story has acceptance criteria that can be turned directly into test cases.

ID	User Story	Acceptance Criteria
Candidate		
US-C1	As a Candidate, I want to ask about open roles and their requirements so that I can decide which roles to apply for.	• Answers come from the official job descriptions. • The assistant distinguishes similar roles (e.g., Backend Developer vs. Data Engineer). • If information is unavailable, the assistant says so.
US-C2	As a Candidate, I want to submit my résumé for a role so that I can be considered for it.	• PDF and DOCX files are accepted. • The candidate receives a confirmation. • An unreadable or incomplete file produces a clear message asking for a new upload.
US-C3	As a Candidate, I want to check the status of my application so that I know where I stand.	• Only the candidate's own status is shown (e.g., Applied, Under Review, Interview Scheduled). • Match scores, internal notes, and other candidates are never shown.
US-C4	As a Candidate, I want to pick or confirm an interview slot so that the interview fits my availability.	• Proposed slots are shown in the chat. • Confirming a slot updates the status and the recruiter's view. • The candidate can request a different slot.

ID	User Story	Acceptance Criteria
US-C5	As a Candidate, I want my personal data kept private so that only authorized HR staff can see my résumé.	• A Candidate account cannot open any HR view or another candidate's data (verified by test).
HR Recruiter		
US-R1	As an HR Recruiter, I want to create and edit job roles with their requirements so that screening is done against a clearly defined role.	• A role has a title, description, required skills, and experience level. • Changes apply to later screening and to candidate answers.
US-R2	As an HR Recruiter, I want to screen résumés against a selected role so that I can evaluate candidates faster.	• Each résumé receives a match score for the selected role. • Incomplete or malformed résumés are flagged with a reason, not silently scored.
US-R3	As an HR Recruiter, I want to match a candidate to open roles, and see ranked candidates for a role, so that I can place people in the best-fit position.	• Given a candidate, the system lists open roles ranked by fit. • Given a role, the system lists candidates ranked by fit.
US-R4	As an HR Recruiter, I want to see why a candidate received a score so that I can trust and verify the result.	• The score comes with matched requirements, missing requirements, and a short explanation.
US-R5	As an HR Recruiter, I want to make the final decision and override the system's recommendation so that a human stays accountable.	• The system never rejects a candidate automatically. • Every override is recorded with who, when, and why.
US-R6	As an HR Recruiter, I want to track each candidate's status so that I can monitor progress.	• Statuses follow a defined workflow. • Status history is kept with timestamps and persists between sessions.
US-R7	As an HR Recruiter, I want the system to help schedule interviews so that I avoid manual coordination.	• The system proposes free slots and prevents double-booking. • The status updates automatically once the candidate confirms.
US-R8	As an HR Recruiter, I want a dashboard of metrics and the candidate pipeline so that I can monitor recruitment at a glance.	• Shows candidates per stage, open roles, time-to-hire, and screening time. • Data matches the underlying records.
Current Employee		
US-E1	As a Current Employee, I want to ask HR policy questions so that I understand company policies and procedures.	• Answers are based only on official HR documents. • Each answer names its source document or section.
US-E2	As a Current Employee, I want to be told when no answer is found so that I am not misled.	• When retrieval finds nothing relevant, the assistant says so and suggests contacting HR. • The assistant does not invent policy.
HR Manager		
US-M1	As an HR Manager, I want to review the audit log of screening decisions so that the process is accountable and fair.	• Each decision shows candidate, role, score, explanation, time, and any override. • The log is visible to HR Managers only.
US-M2	As an HR Manager, I want to upload and update HR policy and job documents so that answers stay current.	• New or updated documents become searchable after upload. • Candidates cannot access internal HR policy documents.

3.2.2 Use Case Summary

ID	Use Case	Primary Actor	Description	Related FR
UC-01	Log In and Access Role-Specific View	All users	User authenticates and is given only the features and data permitted for his or her role.	FR-01, FR-02
UC-02	Manage Job Roles	HR Recruiter	Creates, edits, or closes job roles and their requirements.	FR-03
UC-03	Submit Résumé / Apply	Candidate	Uploads a résumé for an open role and receives confirmation.	FR-04, FR-05
UC-04	Screen Résumé	HR Recruiter	Scores one or more résumés against a selected job role.	FR-05 – FR-10
UC-05	Match Candidate to Roles	HR Recruiter	Ranks open roles for a candidate, or ranks candidates for a role.	FR-07, FR-08
UC-06	Review Match Score Explanation	HR Recruiter	Reviews matched and missing requirements behind a score, then confirms or overrides.	FR-08, FR-10
UC-07	Track Candidate Status	HR Recruiter	Views and updates the status and history of a candidate.	FR-11
UC-08	Schedule Interview	HR Recruiter	Proposes slots to a shortlisted candidate and books the confirmed one.	FR-12, FR-13
UC-09	Ask Recruitment Questions	Candidate	Asks the chat assistant about roles, the hiring process, or his or her own application.	FR-14, FR-16, FR-17
UC-10	Check Application Status	Candidate	Views the status of his or her own application through chat.	FR-11, FR-14
UC-11	Confirm Interview Slot	Candidate	Chooses one of the proposed slots or requests another.	FR-12
UC-12	Ask HR Policy Question	Current Employee	Asks a policy question and receives a sourced answer from official HR documents.	FR-15, FR-17
UC-13	View Recruitment Dashboard	HR Recruiter	Views metrics and the pipeline visualization.	FR-19, FR-20
UC-14	Review Audit Log	HR Manager	Reviews the log of screening decisions and overrides.	FR-21
UC-15	Manage Knowledge Documents	HR Manager	Uploads or updates HR policy and job documents used for retrieval.	FR-22

3.2.3 Key Use Case Scenarios

UC-04: Screen Résumé

Field	Details
Actor	HR Recruiter
Preconditions	• Recruiter is logged in. • At least one open job role with requirements exists. • One or more résumés have been submitted or uploaded.

Field	Details
Main flow	1. Recruiter selects a job role and one or more résumés. 2. System parses each résumé and masks personal attributes unrelated to the job. 3. System compares the résumé with the role requirements and calculates a match score. 4. System shows the score with matched requirements, missing requirements, and a short explanation. 5. System writes the screening result to the audit log. 6. Recruiter confirms or overrides the recommendation; the candidate status is updated.
Alternative flows	A1. Résumé is unreadable or incomplete: the system flags it with the reason and does not assign a score. A2. The role has no requirements defined: the system asks the recruiter to complete the role first.
Postconditions	Screening result is stored, logged, and the candidate status is “Screened”.

UC-08: Schedule Interview

Field	Details
Actors	HR Recruiter (primary); Candidate (secondary)
Preconditions	• Candidate is shortlisted. • Interviewer availability slots exist in the system.
Main flow	1. Recruiter selects a shortlisted candidate and requests scheduling. 2. System proposes available slots. 3. Recruiter approves the slots to offer. 4. Candidate sees the slots in the chat and picks one. 5. System confirms the booking, blocks the slot, and sets the status to “Interview Scheduled”. 6. Recruiter and candidate can both see the confirmed details.
Alternative flows	A1. No slot suits the candidate: candidate requests other slots and the system proposes new ones. A2. Candidate does not respond: status stays “Awaiting Candidate Response” and the recruiter sees it on the dashboard.
Postconditions	Interview is booked with no double-booking and the status history is updated.

UC-09: Ask Recruitment Questions (Candidate Chat)

Field	Details
Actor	Candidate
Preconditions	Candidate is logged in with a Candidate account.
Main flow	1. Candidate asks a question, e.g., “What are the requirements for the Data Analyst role?” or “What is the status of my application?”. 2. System identifies the intent and retrieves information only from permitted sources: public job descriptions, hiring-process information, and the candidate’s own application record. 3. System replies in the chat.
Alternative flows	A1. The question asks for restricted information (match scores, other candidates, internal HR policies, hiring decisions in progress): the system politely declines and explains that it cannot share it. A2. No reliable answer is found: the system says so and suggests contacting HR. A3. The question could refer to several similar roles: the system asks which role is meant.
Postconditions	No restricted data is disclosed; the interaction is recorded.

UC-12: Ask HR Policy Question

Field	Details
Actor	Current Employee
Preconditions	• Employee is logged in. • Official HR documents have been uploaded and indexed.
Main flow	1. Employee asks a policy question, e.g., “How do I request annual leave?”. 2. System retrieves the relevant passages from the official HR documents. 3. System answers using only the retrieved passages and shows the source document and section.
Alternative flows	A1. No relevant passage found: the system says it could not find an answer and suggests contacting HR. A2. Question is ambiguous: the system asks a clarifying question.
Postconditions	Employee receives a grounded answer with its source, or an honest “not found” response.

3.3 Functional Requirements

Priority: Must = required for the capstone deliverables; Should = valuable if time allows. **Pillar:** the capstone pillar the requirement maps to (Agent = Agent System, RAG = RAG Integration, UI = UI Dashboard, Deploy = Deployment).

Requirement	Description	Pillar	Priority
A. Access and Roles			
FR-01 User Authentication	The system shall require users to log in and shall assign each user one role: Candidate, Current Employee, HR Recruiter, or HR Manager.	Deploy	Must
FR-02 Role-Based Access Control	The system shall restrict features and data according to the user's role, as defined in the Role Access Overview (3.1.2), on the server side and not only in the interface.	Deploy / UI	Must
B. Job Roles and Résumé Screening			
FR-03 Job Role Management	The system shall allow HR staff to create, edit, and close job roles, each with a title, description, required skills, experience level, and qualifications.	UI / RAG	Must
FR-04 Résumé Submission	The system shall allow Candidates, and HR staff on their behalf, to upload résumés (PDF or DOCX) linked to a specific job role.	UI	Must
FR-05 Résumé Parsing and Validation	The system shall extract structured information (skills, experience, education) from résumés and shall flag incomplete or malformed résumés with a clear reason instead of failing.	Agent	Must
FR-06 Résumé Screening	The system shall analyze a résumé against the requirements of a selected job role and produce a match score.	Agent	Must
FR-07 Candidate–Role Matching	The system shall rank open roles for a given candidate, and rank candidates for a given role, based on requirements and qualifications.	Agent	Must
FR-08 Explainable Match Score	The system shall present each score with a per-criterion breakdown, the matched and missing requirements, and a short written rationale.	Agent / UI	Must
FR-09 Bias-Safe Screening	The system shall exclude job-irrelevant personal attributes (e.g., name, gender, age, photo, marital status) from the scoring input.	Agent	Must
FR-10 Human Final Decision	The system shall present scores as recommendations only. It shall not reject or advance a candidate without HR confirmation, and every override shall be recorded.	Agent	Must
C. Recruitment Workflow			
FR-11 Candidate Status Tracking	The system shall maintain each candidate's status (Applied, Screened, Shortlisted, Interview Scheduled, Interviewed, Offered, Rejected, Withdrawn) with a timestamped history that persists across sessions.	Agent	Must
FR-12 Interview Scheduling	The system shall propose available interview slots, let the Candidate confirm or request another slot, prevent double-booking, and update the status automatically.	Agent	Must
FR-13 Multi-Step Workflow Planning	The system shall plan and execute multi-step recruitment workflows (e.g., screen → shortlist → propose slots → update status), showing the plan and each step's outcome, with HR approval at key steps.	Agent	Must
D. Chat Assistants and Knowledge Retrieval			

Requirement	Description	Pillar	Priority
FR-14 Candidate Chat Assistant	The system shall provide a chat interface where Candidates ask about open roles, requirements, the hiring process, their own application status, and interview details, using only candidate-permitted information.	UI / RAG	Must
FR-15 Employee HR Policy Chat	The system shall provide a chat interface where Current Employees ask HR policy questions and receive answers retrieved from official HR documents, with the source shown.	UI / RAG	Must
FR-16 Similar-Role Disambiguation	The system shall distinguish between similar but distinct roles when retrieving job information, and ask for clarification when a question is ambiguous.	RAG	Should
FR-17 Unanswerable and Restricted Questions	When no reliable answer exists, or the question requests restricted information, the system shall say so, shall not fabricate an answer, and shall direct the user to HR where appropriate.	RAG / Agent	Must
FR-18 Successful-Hire Pattern Reference	The system shall use a sample, anonymized archive of past successful hires as an additional reference signal when matching candidates.	RAG	Should
E. Dashboard and Administration			
FR-19 Recruitment Dashboard	The system shall provide an HR-facing dashboard showing open roles, candidates per stage, time-to-hire, and screening-time metrics.	UI	Must
FR-20 Pipeline Visualization	The system shall visualize the candidate pipeline by stage, filterable by job role.	UI	Must
FR-21 Audit Log Viewer	The system shall let HR Managers view the audit log of screening decisions and overrides.	UI	Should
FR-22 Knowledge Document Management	The system shall allow HR Managers to upload, update, and remove HR policy documents, job descriptions, and the hire-pattern archive, and make them available for retrieval.	UI / RAG	Must

3.4 Non-Functional Requirements

Targets marked “proposed” are starting values that the team should confirm or adjust after testing.

Requirement	Description	How It Is Verified
Security and Privacy		
NFR-01 Access Control	The system shall enforce role permissions on every request so that no user can reach data or features outside his or her role.	Test: Candidate and Employee accounts attempt HR actions and are denied.
NFR-02 Data Privacy	The system shall keep candidate data separate from Candidate and Employee views and shall follow the principles of Egypt’s Personal Data Protection Law (No. 151 of 2020). Only sample or anonymized data is used.	Leakage test: zero HR-only or other-candidate data appears in Candidate or Employee views.
NFR-03 Secrets Management	API keys, database credentials, and tokens shall not be hard-coded or committed to the repository.	Repository scan finds no secrets; configuration is loaded from environment variables or a secret store.

Requirement	Description	How It Is Verified
NFR-04 Data Protection	Data in transit shall be encrypted (HTTPS) and uploaded résumés shall be stored with restricted access.	Deployed site uses HTTPS; storage permissions reviewed.
NFR-05 Audit Logging	The system shall log every screening decision with the user, candidate, role, score, explanation, timestamp, and any override. Logs shall be append-only and visible only to HR Managers.	Each test screening produces a log entry; logs cannot be edited from the interface.
Accuracy, Fairness, and Efficiency		
NFR-06 Matching Accuracy	The system shall rank candidates correctly for a defined role in at least 8 of 10 representative test cases.	Ground-truth ranking prepared by the team for 10 test cases; compare with system output.
NFR-07 HR Policy Answer Accuracy	The system shall answer representative HR policy questions correctly and in line with the official documents (proposed: at least 8 of 10).	Question set with expected answers and sources; check correctness and cited source.
NFR-08 Bias Mitigation	The matching logic shall avoid at least one named bias risk (e.g., gender or name bias) and the approach shall be documented in a short write-up.	Test: identical résumés differing only in name or gender receive the same score; write-up included.
NFR-09 Screening Efficiency	The system shall measurably reduce manual résumé screening time compared with a fully manual baseline (proposed target: at least 50%).	Time a fixed batch of résumés manually and with the system; report the difference.
Performance, Reliability, and Usability		
NFR-10 Response Time	Chat answers shall be returned within 10 seconds for 90% of queries and a single résumé screening within 30 seconds (proposed).	Measure response times over a set of test queries.
NFR-11 Error Handling and Reliability	The system shall handle incomplete or malformed résumés and failures of external AI services without crashing or losing data, and shall show a clear error message.	Test with corrupted, empty, and unsupported files and with the AI service unavailable.
NFR-12 Availability	The deployed system shall be accessible through a public link during the evaluation and demo period.	Deployment link checked before the final presentation.
NFR-13 Usability	Chat and dashboard shall be usable without training, display role-appropriate views, and work on common desktop browsers.	Informal user test with team members or classmates in each role.
NFR-14 Language Support	The system should handle résumés and chat questions in English and, if the team agrees, Arabic (proposed: Should).	Test queries and sample résumés in each supported language.
Maintainability and Deployment		
NFR-15 Deployability and Maintainability	The system shall be modular, configurable through environment settings, and documented with a README covering installation and execution.	A teammate follows the README from scratch and runs the system.

3.5 Assumptions, Constraints, and Scope Boundaries

- **Human in the loop:** TawzeefAI recommends; HR decides. The system never rejects candidates automatically.
- **Sample data only:** all résumés, job descriptions, HR policies, and the hire-pattern archive are sample or anonymized documents.

- **Scheduling:** interview slots come from availability stored inside the system. Integration with external calendars or sending real emails is out of scope unless the team decides to add it.
- **Candidate scope:** candidates can only access public job information, their own application, and interview scheduling. They cannot see scores, other candidates, or internal HR policies.
- **Out of scope:** payroll, offer-letter generation, performance reviews, and legal advice.
- **To confirm with the lecturer or team:** whether Arabic is supported (NFR-14), the exact status workflow (FR-11), and the target values proposed in NFR-07, NFR-09, and NFR-10.